



Examiners' Report Summer 2026

Pearson Edexcel GCE

In Further Mathematics (9FM0)

Paper 4C Further Mechanics 2

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General Comments

Performance on this paper was generally positive. Learners demonstrated a secure grasp of principles from both static and dynamic mechanics, with many responses showing confidence in algebraic integration and in working with logarithmic and exponential structures in calculus-based mechanics contexts.

Stronger responses were characterised by clear, well-structured derivations in “show that” questions and by the accurate construction of equations of motion for multi-stage systems.

Areas for improvement include maintaining dimensional consistency when manipulating variable mass systems and taking care when setting up geometric components in more complex physical layouts. Learners should also ensure that all relevant external forces, particularly gravity, are included when resolving forces or constructing equations of motion. Where a proof requires an explicit conclusion or a specific form of acceleration, this should be stated clearly.

Comments on individual questions

Question 1

Part (a)

Part (a) was completed successfully by the majority of learners, who showed a good understanding of resolving forces vertically. Where marks were not awarded, this was usually because responses did not resolve the tension component along the vertical axis.

Part (b)

Part (b) was generally well answered, with learners setting up the correct horizontal equation of motion involving angular velocity. However, a small number of responses started with the claim that $T = mr\omega^2$ and did not justify why or how the trigonometric ratios had cancelled out from both sides of their equations to reach the printed result.

Part (c)

Part (c) was answered well. The few responses who did not gain full marks typically made an arithmetic slip when multiplying the period expression $\frac{2\pi}{\omega}$ by 5.

Question 2

Part (a)

Part (a) generated many strong responses, although this “show that” question was more challenging for a notable minority of learners. Dimensional inconsistency was a common reason why marks were not awarded.

Some responses did not consider all three parts of the composite shape together. For example, they either added the new glass cone without removing the wooden core or removed only the wooden part. Slips in using the standard distance formula for the centre of mass of a cone were also seen.

Learners should also ensure that any change from a negative measurement relative to 0 to a positive distance d is justified explicitly.

Part (b)

Part (b) was more challenging. Some responses used only r in the numerator of the trigonometric ratio, while others used d in the denominator rather than the correct geometric distance $2r - d$.

Question 3

Part (a)

Part (a) was generally well answered, with most responses forming the differential equation correctly. Some responses found the subsequent integration more challenging. Although the chain rule was often recognised, fractional manipulation was a common source of error.

Part (b)

Part (b) was generally well attempted, with many learners converting directly to logarithmic form. Responses that used $\operatorname{arcsinh}$ usually recognised the need to convert the result into logarithms.

A common misinterpretation was to integrate the final expression from part (a). Some otherwise good responses also spent more time than necessary justifying the rearrangement of the logarithmic form into the required answer.

Question 4

This centre of mass question was answered well, with many responses showing a secure understanding of the required methods.

Part (a)

Part (a) was completed successfully by most learners. Where marks were not awarded, this was usually because responses used an incorrect distance for the centre of mass of at least one rod, with BE being the most common source of error. A small number of responses also omitted the mass of rod CD from the total framework mass calculation.

Part (b)

In part (b), almost all learners found the total mass successfully. Some responses found the variable density function more challenging, with errors seen when integrating $x^2(10 - 3x)$. Most responses then proceeded accurately through the remaining integration steps to reach the correct solution.

Question 5

This circular motion question was more challenging for many learners, although the initial set-up was accessible.

Part (a)

Part (a) was completed successfully by almost all learners. As this was a printed answer, learners had the opportunity to review and refine their approach. Where marks were not awarded, this was usually because responses did not apply energy principles, or because sign errors in the initial equations led to work that was inconsistent with the given result.

Part (b)

Part (b) was generally well answered. Where marks were not awarded, this was usually because responses either omitted the gravitational force or did not resolve it correctly along the radial axis.

Part (c)

Part (c) was more challenging. Many responses calculated only the radial component of the acceleration. Some responses attempted to use Pythagoras' theorem directly to combine the radial acceleration with the vertical component mg . Responses that identified both distinct acceleration components were usually able to combine them correctly to reach the final answer.

Question 6

This calculus-based centre of mass question was accessible in its initial stages, although the final part was more demanding.

Part (a)

The “show that” structure in part (a) gave learners the opportunity to review and refine their working. Most responses reached the correct result, supported by straightforward integrals and logarithmic terms. Where marks were not awarded, this was often because responses attempted to integrate incorrect structural forms, such as y^2 or x^2y .

Part (b)

In part (b), most learners made a valid attempt to find $\bar{y}$ but some were unable to progress further. Among responses that established a correct method for the moments equation, a common error was to use $\bar{x}$ directly from part (a), rather than recognising that the physical set-up required 2 to be subtracted from it.

Even where the problem was set up correctly, the numerical manipulation was demanding, and some responses did not reach the final mark. The stronger responses used efficient and accurate methods throughout.

Question 7

This multi-stage simple harmonic motion question was generally well attempted, with many correct responses seen.

Part (a)

In part (a), most learners achieved the correct result. Where marks were not awarded, this was usually because responses contained algebraic slips or omitted terms and letters within Hooke's law formulae.

Part (b)

The most common error in part (b) was the omission of the gravitational force from the dynamic equation. Some responses also changed distances or lengths repeatedly in an attempt to obtain the printed period. The most efficient solutions used l and $0.6l$ from the outset.

Some otherwise correct responses did not gain full marks because they omitted a concluding statement explicitly identifying simple harmonic motion, or did not state the acceleration in its correct differential form.

Part (c)

Part (c) was answered well by most learners. Where marks were not awarded, this was usually because responses did not use $0.4l$ for the amplitude, or because the angular frequency ω was confused with the total time period.

Part (d)

Part (d) was more challenging. Some responses calculated a relevant time interval but did not convert this into the required final solution. This was usually due to calculating the wrong total time interval, rather than to difficulties manipulating the surds.

A further common error was to use a distance of $0.3l$ rather than of the correct value of $0.1l$ in their kinematic equations.